RAJALAKSHMI ENGINEERING COLLEGE

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K2

Started on	Wednesday, 19 August 2026, 9:04 AM
State	Finished
Completed on	Wednesday, 19 August 2026, 10:50 AM
Time taken	1 hour 45 mins
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Given an array `nums` of size `n`, return *the majority element*.

The majority element is the element that appears more than `⌊n / 2⌋` times. You may assume that the majority element always exists in the array.

Example 1:

```
Input: nums = [3,2,3]
Output: 3
```

Example 2:

```
Input: nums = [2,2,1,1,1,2,2]
Output: 2
```

Constraints:

- `n == nums.length`
- `1 <= n <= 5 * 10^4`
- `-2^31 <= nums[i] <= 2^31 - 1`

For example:

Input	Result
3 3 2 3	3
7 2 2 1 1 1 2 2	2

Answer: (penalty regime: 0 %)

```
#include<stdio.h>
int countInRange(int arr[],int left,int right, int candidate){
    int count =0;
    for(int i=left;i<=right;i++){
        if(arr[i]==candidate) count++;
    }
    return count;
}
int majorityRec(int arr[],int left,int right){
    if(left==right){return arr[left];}
    int mid =(left+right)/2;
    int leftM=majorityRec(arr,left,mid);
    int rightM = majorityRec(arr,mid+1,right);
    if(leftM==rightM){return leftM;}
    int leftC=countInRange(arr,left,right,leftM);
    int rightC=countInRange(arr,left,right,rightM);

    return (leftC>rightC)? leftM:rightM;
}

int majorityElement(int arr[],int n){
    return majorityRec(arr,0,n-1);
}

int main(){
    int m;
```

```
27      scanf("%d",&m);
28      int arr[m];
29      for(int i=0;i<m;i++){
30          scanf("%d",&arr[i]);
31      }
32      printf("%d",majorityElement(arr,m));
33      return 0;
34  }
```

	Input	Expected	Got	
✔	3 3 2 3	3	3	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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